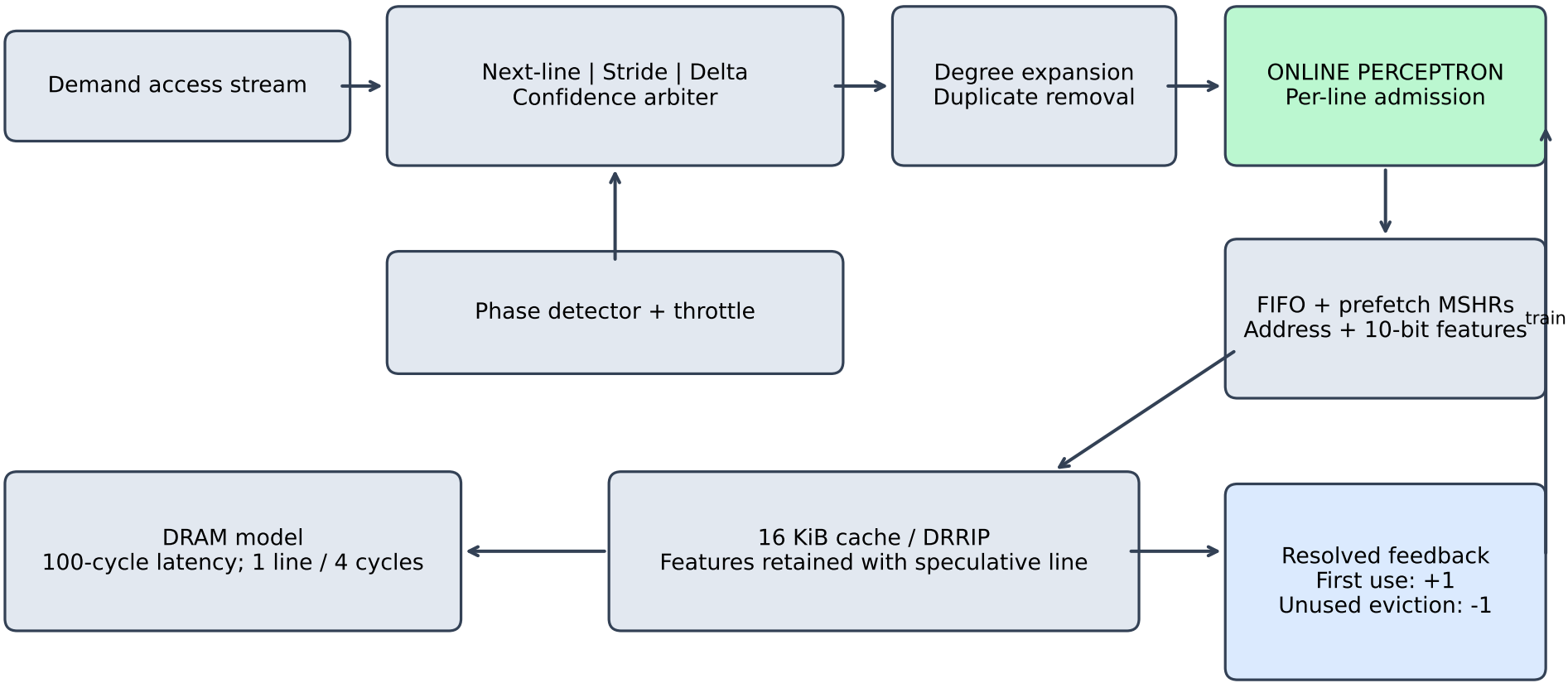
# Optional online ML admission filter in the existing cache architecture




score = bias + sum(weight × bipolar feature) | saturating online margin update

Green block is new. Metadata follows the actual line; software phase labels never enter the learner.